

Monte Carlo simulation of π^- interactions on a polyethylene target and optimization of the target thickness

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Introduction

- Monte Carlo simulations are a useful tool in experimental high-energy physics, where detector design and analysis strategies must be optimized before data taking.
- By generating synthetic events according to known physical laws and detector responses, Monte Carlo methods allow to study the experimental reconstruction and its statistical performance in a controlled environment
- In this work, a Monte Carlo simulation of a π^- beam with an energy of 100 GeV impinging on a polyethylene target is developed.
- The signal processes $\pi^- p \rightarrow \pi^0 n$ and $\pi^- p \rightarrow \eta n$, followed by the decay $\pi^0 \rightarrow \gamma\gamma$ and $\eta \rightarrow \gamma\gamma$, are considered.
- The main goal of the study is the **optimization of the target thickness L** , in order to **maximize the statistical significance** of the reconstructed signal while preserving a good **invariant-mass resolution**.

Experimental setup

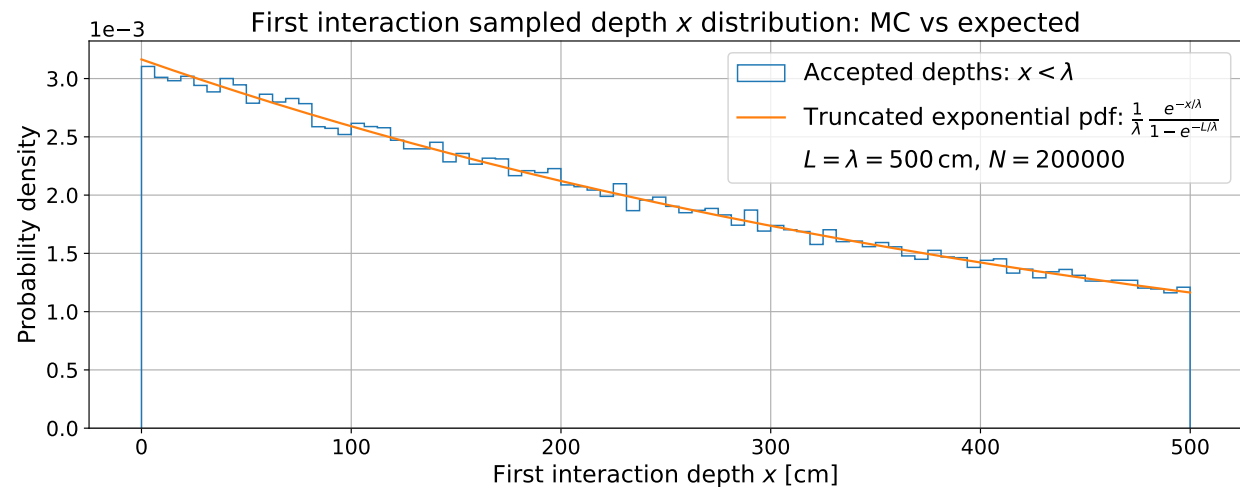
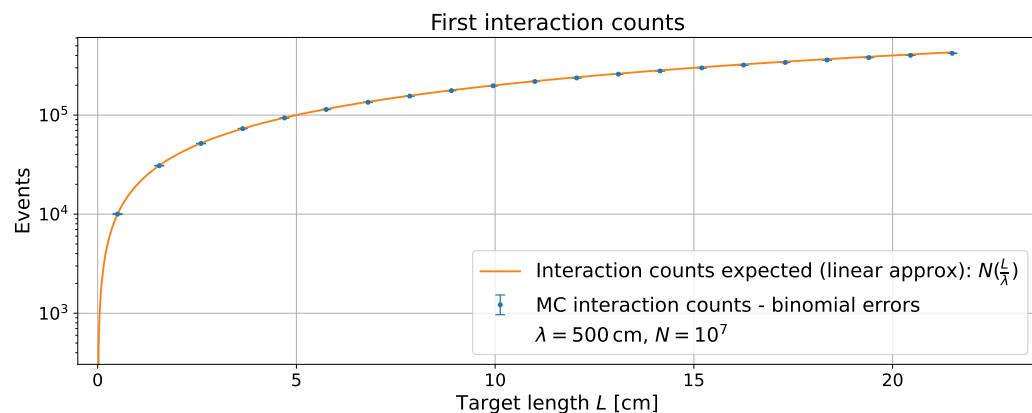
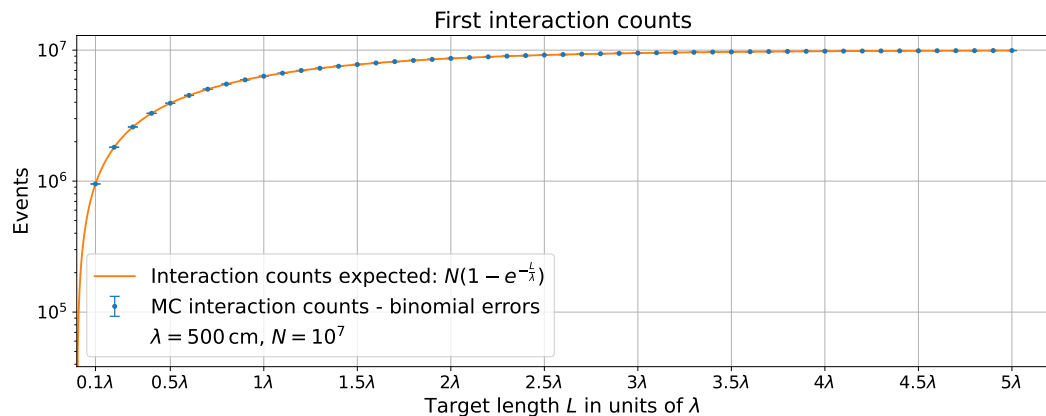
- **Incident beam** → 100Gev nominal, no beam divergence, single relevant interaction per incident pion (neglects secondary hadronic interactions inside the target), no pile-up effects
- **Target** → CH_2 , homogeneous medium, only hydrogen protons at rest considered for the interaction
- **Signal vs background** → only $\pi^0/\eta \rightarrow \gamma\gamma$ process are signal contributions. All other interactions are treated as background
- **Detector** → sensitive exclusively to γ final states. No information on the recoiling neutron or on the primary interaction vertex. The optimization procedure is entirely based on the reconstruction of the invariant mass $m_{\gamma\gamma}$
- **Target and detector geometry**
 - Target** → cylinder of $R = 10$ cm, length L , axis aligned along beam direction
 - Detector** → circular surface of $R = 100$ cm, centered on beam axis, located at $z = 10$ m from target

Interaction depth sampling

- **Probability density** for the first interaction to occur at depth $z \longrightarrow P(z) = \frac{1}{\lambda} \exp(-\frac{z}{\lambda})$
- **Interaction length:** $\lambda = \frac{1}{n \sigma_{tot}} \approx 499.95 \text{ cm}$ where $\longrightarrow n_p = 2 \frac{p}{M} N_A \approx 8.00 \cdot 10^{22} \text{ cm}^{-3}$
 $\longrightarrow \sigma_{tot}(\pi^- p, 100 \text{ GeV}) \approx 25 \text{ mb}$
- The **probability** for a pion to undergo **at least one interaction** within a target of thickness L :
 $\longrightarrow P_{inter}(L) = \int_0^L \frac{1}{\lambda} e^{-x/\lambda} = 1 - e^{-L/\lambda} ; \text{ If } L \ll \lambda \longrightarrow P_{inter}(L) \approx \frac{L}{\lambda}$
- Approximation: z is therefore used as the **interaction position** along the target: $L_{meson} \simeq \frac{E}{m} c \tau \ll \lambda$
- **z sampling** from **inverse transform method**:
 $\longrightarrow$ the CDF $F(z) = P(Z \leq z)$ is uniformly distributed in $[0,1]$
 $\longrightarrow u = F(z) = 1 - e^{-z/\lambda} \longrightarrow \boxed{z = -\lambda \ln u}$

Interaction depth sampling validation

Monte Carlo vs analytical prediction for the **number of first interactions** vs target thickness L .



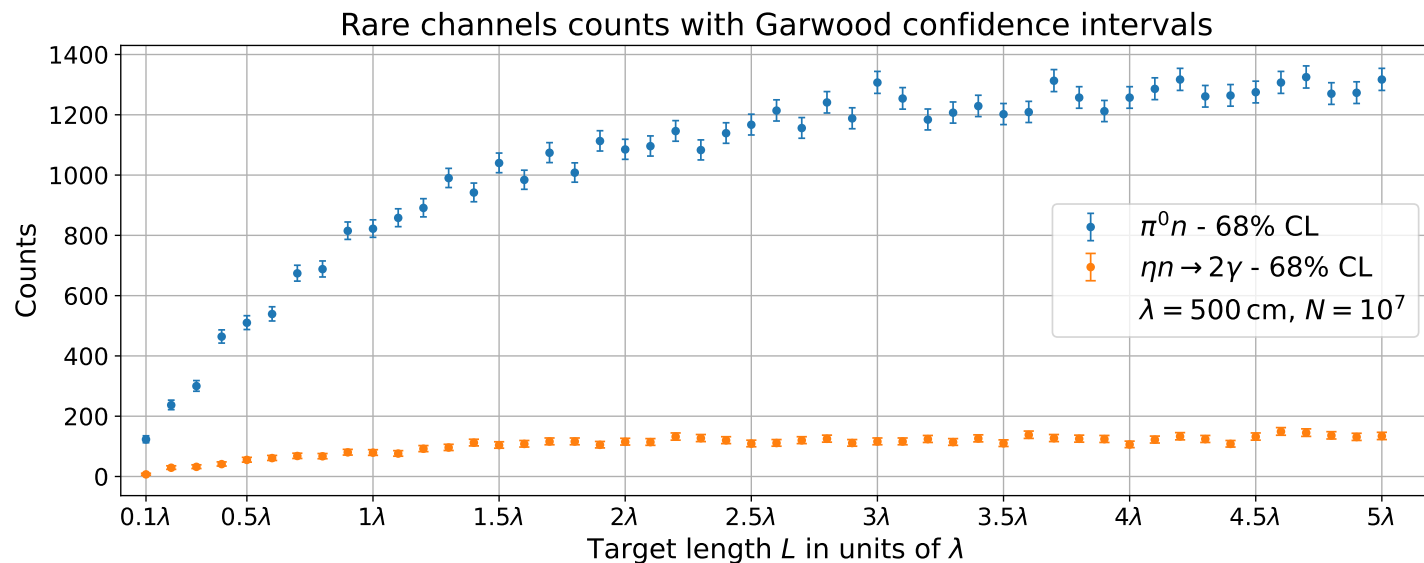
Distribution of the **sampled interaction depth z** for a fixed target thickness, compared to the corresponding **truncated exponential probability density**.

Small deviations are due to finite Monte Carlo statistics and binning effects.

Channel probabilities

- Given an interaction inside the target, the final state of the event is selected probabilistically according to the **relative production cross sections** of the available channels

$$p_{\eta} = \frac{\sigma_{2\gamma}(\pi^{-} p \rightarrow \eta n)}{\sigma_{tot}(\pi^{-} p, 100 \text{ GeV})} \approx 0.001\% \quad ; \quad p_{\pi^0} = \frac{\sigma_{tot}(\pi^{-} p \rightarrow \pi^0 n)}{\sigma_{tot}(\pi^{-} p, 100 \text{ GeV})} \approx 0.013\% \quad ; \quad p_{bkg} = 1 - p_{\eta} - p_{\pi^0} = 99.9\%$$



Expected event counts for the rare $\pi^{-} p \rightarrow \pi^0 n$ and $\pi^{-} p \rightarrow \eta n$ channels vs target thickness L . Error bars represent 68% confidence intervals computed using the **Garwood construction** for Poisson distributed counts.

Meson angular distribution

- In the process $\pi^- p \rightarrow X n$ the information on the angular distribution in the center-of-mass frame is encoded in the **differential cross section** $d\sigma/d\tau$ with $\tau \equiv -t$
- From tabulated values of $d\sigma/d\tau \rightarrow \Delta\sigma_i \simeq y_i \tau_i$ with $\Delta\sigma_i$ corresponding to the bin area in the $(\tau, d\sigma/d\tau)$ plane

$\left\{ \begin{array}{l} y_i \equiv \left(\frac{d\sigma_i}{d\tau_i} \right) \rightarrow \text{Differential cross-section} \\ \tau_i \rightarrow \text{Bin width} \end{array} \right.$

constant within each bin
- The **probability** for an event **to fall into bin i** is defined as the fraction of the total cross section carried by that bin

$$\rightarrow P_i = \frac{\Delta\sigma_i}{\sum_{j=1}^N \Delta\sigma_j} \quad \text{with} \quad \sum_{i=1}^N P_i = 1$$

- The resulting **probability density function for τ** is piecewise constant

$$p(\tau) = \begin{cases} \frac{P_i}{\Delta\tau_i} = \frac{y_i}{\sum_j y_j \Delta\tau_j} & \tau \in [\tau_i^{\text{low}}, \tau_i^{\text{high}}], \\ 0 & \text{otherwise} \end{cases}$$

Meson angular distribution

- **Discrete CDF** from the bin probabilities $\longrightarrow CDF_k = \sum_{i=1}^k P_i$; $CDF_N = 1$

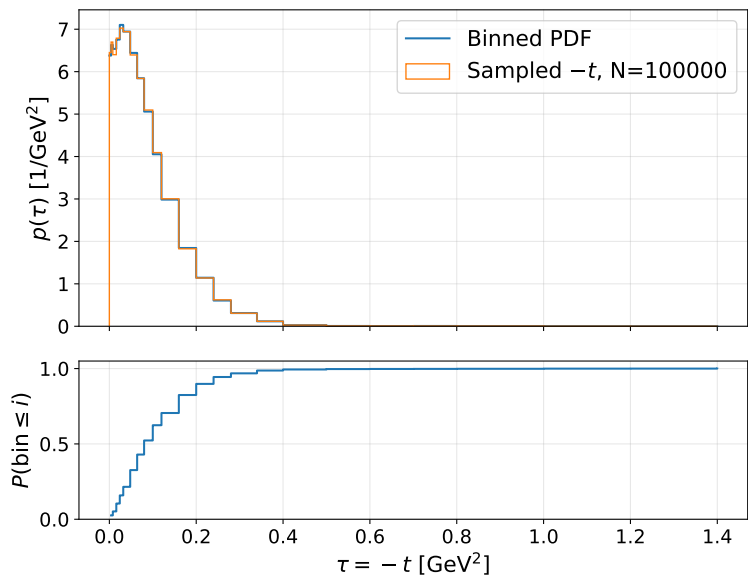
The value CDF_k represents the **probability** that an event **belongs to a bin with index $\leq k$**

- **$\tau \equiv -t$ sampling:**

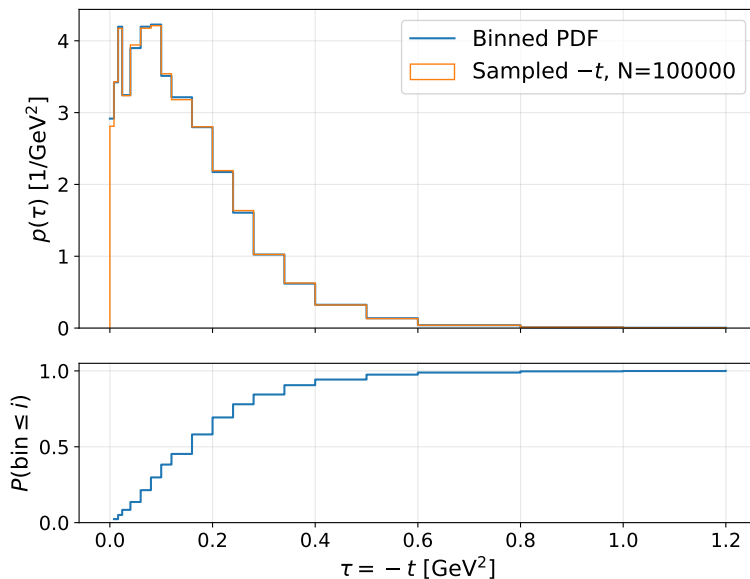
$\longrightarrow u \sim \mathcal{U}[0,1]$; the first bin satisfying $CDF_i > u$ is selected. This assigns bin i with P_i

$\longrightarrow \tau \sim \mathcal{U}[\Delta\tau_i]$

Binned pdf and CDF for t-sampling from $\frac{d\sigma_{\pi^0}}{dt}$



Binned pdf and CDF for t-sampling from $\frac{d\sigma_{\eta}}{dt}$



Piecewise-constant pdf and CDF reconstructed from the tabulated $d\sigma/d\tau$, together with Monte Carlo generated samples.

The strong concentration of events at **small values of τ** exhibits the **forward nature** of the meson production at high energy

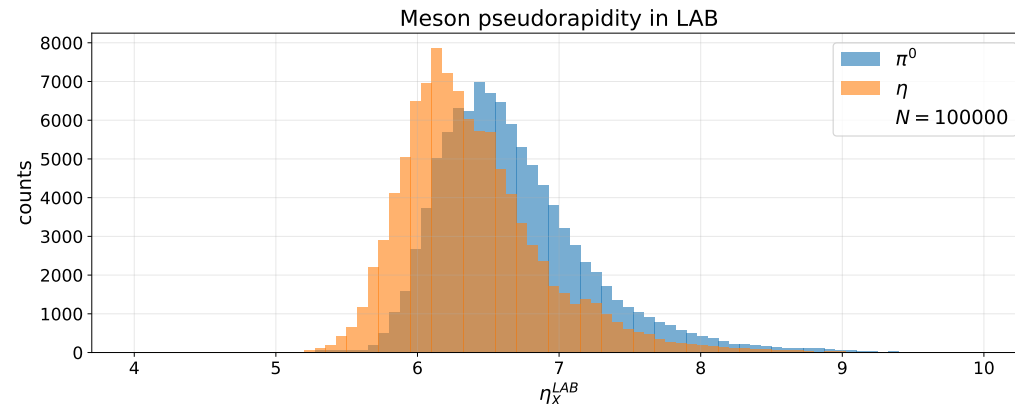
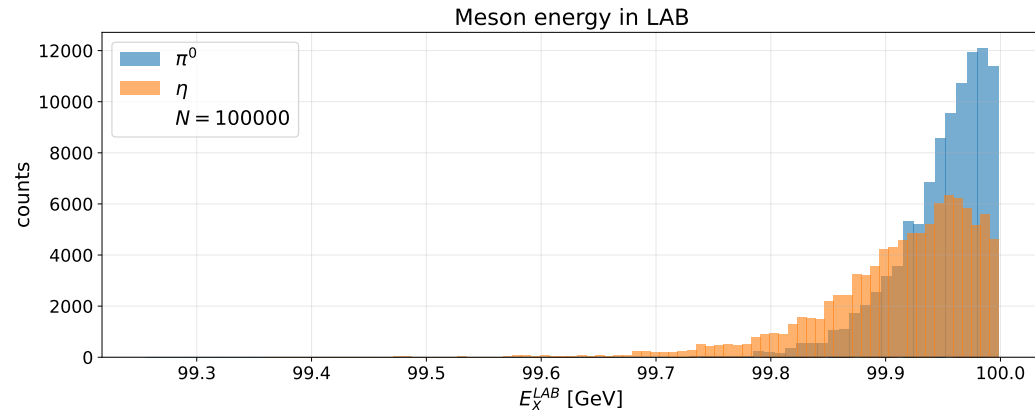
Event kinematics

Conventions:

- **Polar angle θ** $\longrightarrow$ respect to the z axis: the direction of the incoming beam
- **Azimuthal angle ϕ** $\longrightarrow$ respect to the transverse (x,y) plane around the beam axis
- **Pseudorapidity η** $\longrightarrow$ $\eta = -\ln \tan(\theta/2)$

Two body scattering: $\pi^- p \rightarrow X n$

- **Lab frame:** $s = (P_p + P_\pi)^2 = m_p^2 + m_\pi^2 + 2m_p E_\pi$; $\beta_{CM} = \frac{|\vec{P}_{tot}|}{E_{tot}} = \frac{p_\pi}{m_p + E_\pi}$
- **CM frame:** $p^{CM} = \frac{\sqrt{\lambda(s, m_X^2, m_n^2)}}{2\sqrt{s}}$; $\cos \theta^{CM} = \frac{t - m_\pi^2 - m_X^2 + 2E_\pi^{CM} E_X^{CM}}{2p_\pi^{CM} p_X^{CM}}$; $\phi^{CM} \sim U[0, 2\pi]$
- Meson momentum in CM $\longrightarrow \vec{p}_X = p^{CM}(\sin \theta^{CM} \cos \phi^{CM}, \sin \theta^{CM} \sin \phi^{CM}, \cos \theta^{CM})$ $\longrightarrow p_X^{LAB} = \Lambda(\vec{\beta}^{CM}) p^{CM}$
and transformation in LAB



Decay $X \rightarrow \gamma\gamma$

Meson rest frame:

$$E_\gamma^{(X)} = \frac{m_X}{2}$$

$$\cos\theta_\gamma^{(X)} \sim U[-1, 1]$$

$$\phi_\gamma^{(X)} \sim U[0, 2\pi]$$

$$k_{1,2}^{(X)} = \left(\frac{m_X}{2}, \pm \frac{m_X}{2} \cdot \hat{n} \right)$$

$$\hat{n} = (\sin\theta_\gamma^{(X)} \cos\phi_\gamma^{(X)}, \sin\theta_\gamma^{(X)} \sin\phi_\gamma^{(X)}, \cos\theta_\gamma^{(X)})$$

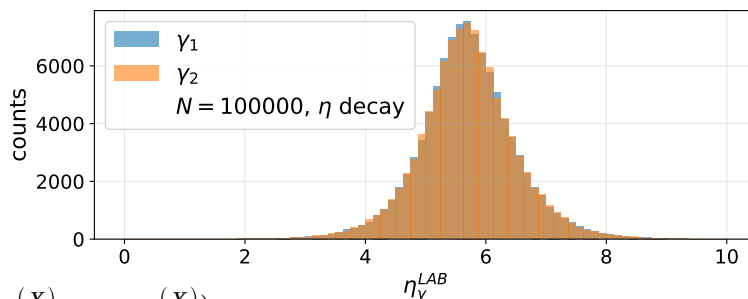
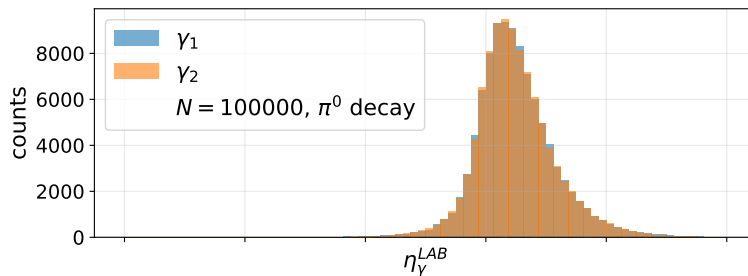
LAB frame:

$$\vec{\beta}_X = \frac{\vec{p}_X^{LAB}}{E_X^{LAB}} \longrightarrow E_\gamma^{LAB} = \gamma_X E_\gamma^{(X)} (1 + \vec{\beta}_X \cdot \hat{n})$$

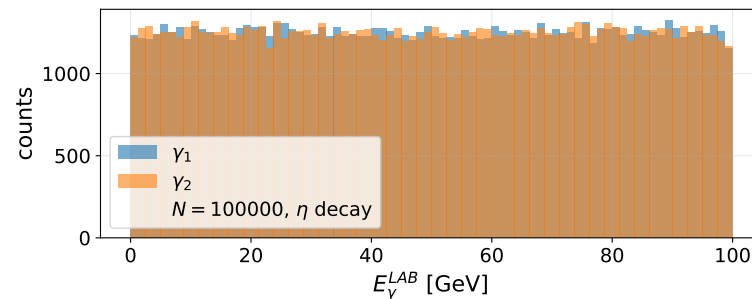
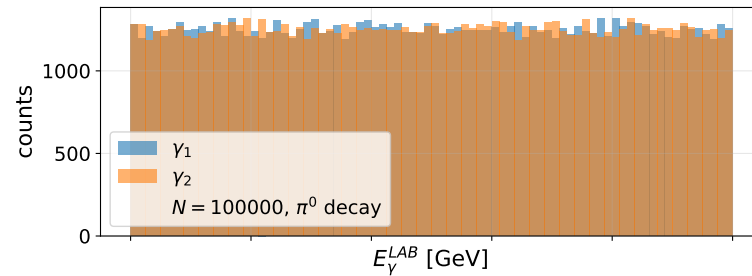
Opening angle in

LAB between the two photons $\longrightarrow \Delta R(\gamma, \gamma) = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$

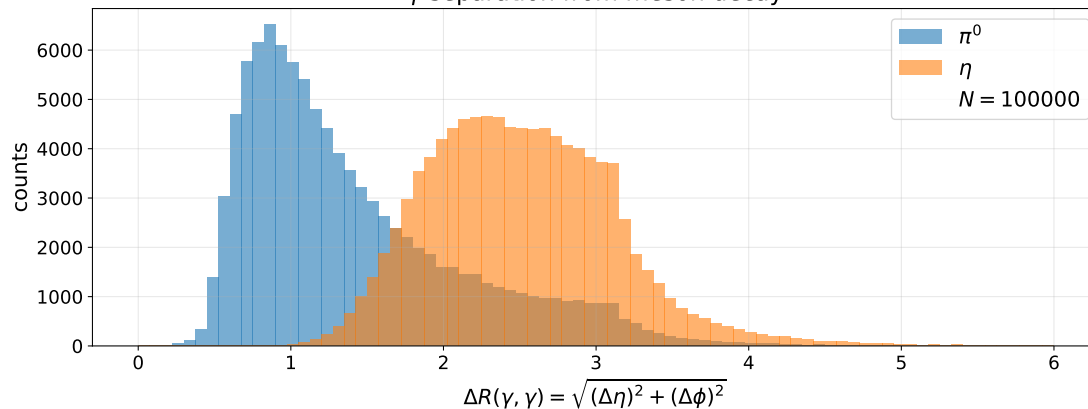
γ pseudorapidity in LAB from meson decay



γ energy in LAB from meson decay



γ separation from meson decay



Detector response

- Geometry acceptance:** $\longrightarrow$ a photon is accepted if its trajectory intersects the detector surface $\longrightarrow A_{geom}(\gamma) = \Theta(R_{cal}^2 - x_\gamma^2 - y_\gamma^2)$
- Photon escape probability:** $P_{escape}(l) = \exp\left(-\frac{l}{\lambda_\gamma}\right)$ with λ_γ approximated in the **electron-positron pair** production regime: $\longrightarrow \lambda_\gamma = \frac{9}{7} X_0 ; X_0 = 50.31 \text{ cm}$
- Energy smearing** $\longrightarrow E_\gamma^{meas} \sim N(E_\gamma^{true}, \sigma_E^2)$
worsening of the resolution induced by photon path length inside the target
 $\longrightarrow \frac{\sigma_E}{E_\gamma^{true}} = \sqrt{\left(\frac{a}{\sqrt{E_\gamma^{true}}}\right)^2 + b_{eff}^2} \longrightarrow b_{eff} = \sqrt{b^2 + \left(k_E \frac{l}{\lambda_\gamma}\right)^2}$
- Position smearing** $\longrightarrow x_\gamma^{meas} \sim N(x_\gamma^{true}, \sigma_x^2)$
worsening of the resolution induced by photon path length inside the target $\longrightarrow \sigma_x = \sigma_{x,0} \left(1 + k \frac{l}{\lambda_\gamma}\right)$
- Energy and angular separation threshold**
 $\longrightarrow E_\gamma^{meas} > E_{threshold} = 0.1 \text{ GeV} \longrightarrow \Delta R^{meas} > \Delta R_{min} = 0.02$

Event reconstruction

- **Invariant mass reconstruction** $\longrightarrow m_{\gamma\gamma}^2 = 2 E_1^{meas} 2 E_2^{meas} (1 - \cos \alpha^{meas})$
with $\cos \alpha^{meas} = \hat{n}_1 \hat{n}_2$ opening angle between the two photons
- **Background sampling** $\longrightarrow$ **log-uniform distribution**, sampled E_γ with **inverse transform method**
 $\longrightarrow \log E_\gamma \sim U(\log E_{min}, \log E_{max}) ; p(E_\gamma) = \frac{1}{E_\gamma \ln(E_{max}/E_{min})} \longrightarrow E_\gamma = E_{min} \left(\frac{E_{max}}{E_{min}} \right)^u ; u \sim U[0, 1]$
- **Iterative fit on signal** $\longrightarrow$ fit exclusively the signal shape with a **Gaussian model** getting μ_{fit}, σ_{fit}
 $\longrightarrow$ iterative fit in the range $\mu_{fit} \pm k\sigma_{fit}$ to get **mass resolution**
- **Oversampling** $\longrightarrow$ extremely small expected number of produced neutral mesons
 $\longrightarrow$ **fixed** number of generated events for signal and background
 $\longrightarrow$ event reconstruction in oversampled sets is dependent from photon escape probability and detector response only, not from channel probabilities
 $\longrightarrow$ To recover physical expected yields **rescale with weights**: $w_{phys} = \frac{N_{expected}}{N_{generated}}$

Event reconstruction

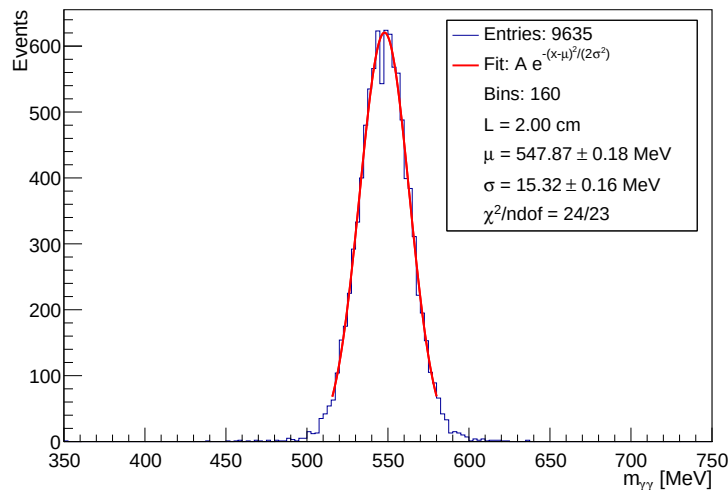
- **Signal and background yields** $\longrightarrow S(L), B(L)$
number of the events in the invariant-mass oversampled spectra in an adaptive window centered at the fitted peak position $\longrightarrow |m_{\gamma\gamma} - \mu(L)| < k \sigma(L)$
- **Uncertainty estimation**
 - $\longrightarrow$ Poisson bootstrap technique: create spectra bootstrap replicas by fluctuating the content of each bin according to a Poisson distribution $n_i^{new} \sim Poiss(n_i)$.
 - $\longrightarrow$ For each bootstrap replica, the full analysis chain is repeated, getting $S(L), B(L)$ and mass resolution
 - $\longrightarrow$ Statistical uncertainty on each observable: standard deviation of its bootstrap distribution
- **Significance**

$$Z(L) = \frac{S(L)}{\sqrt{(S(L) + B(L))}}$$

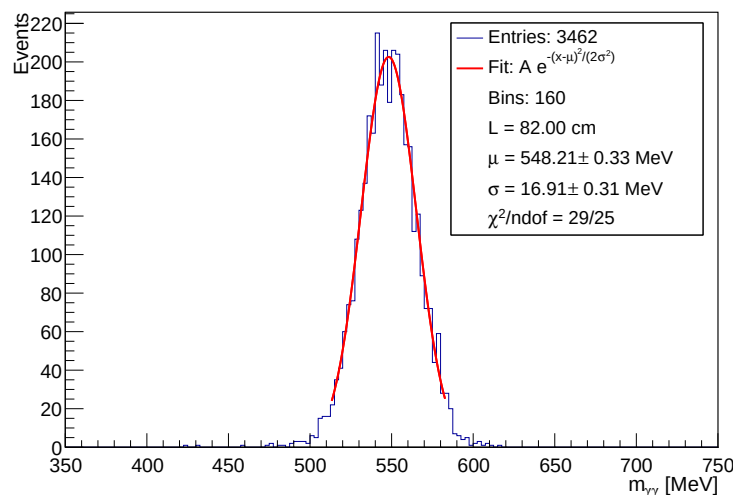
Measures how large the signal yield is compared to the typical statistical fluctuations of the total event count: estimation of the statistical signal detectability from the background

Optimization of the target thickness

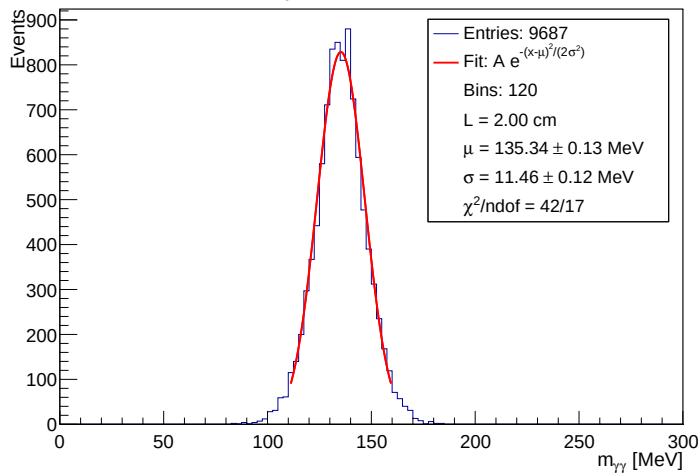
Fit η peak at L=2.00 cm



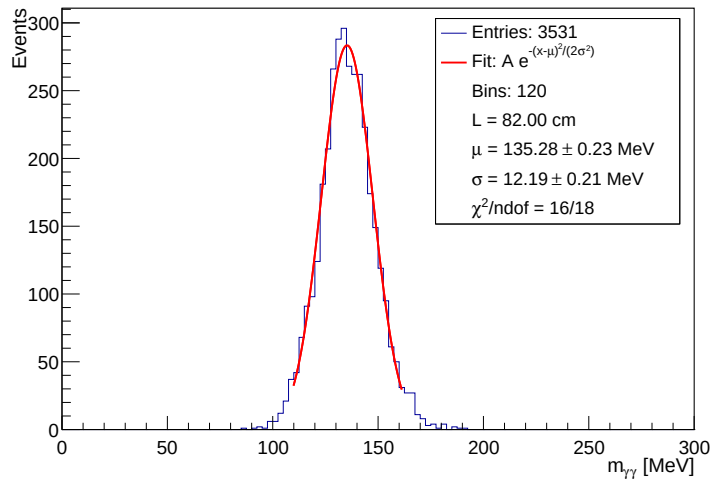
Fit η peak at L=82.00 cm



Fit π^0 peak at L=2.00 cm



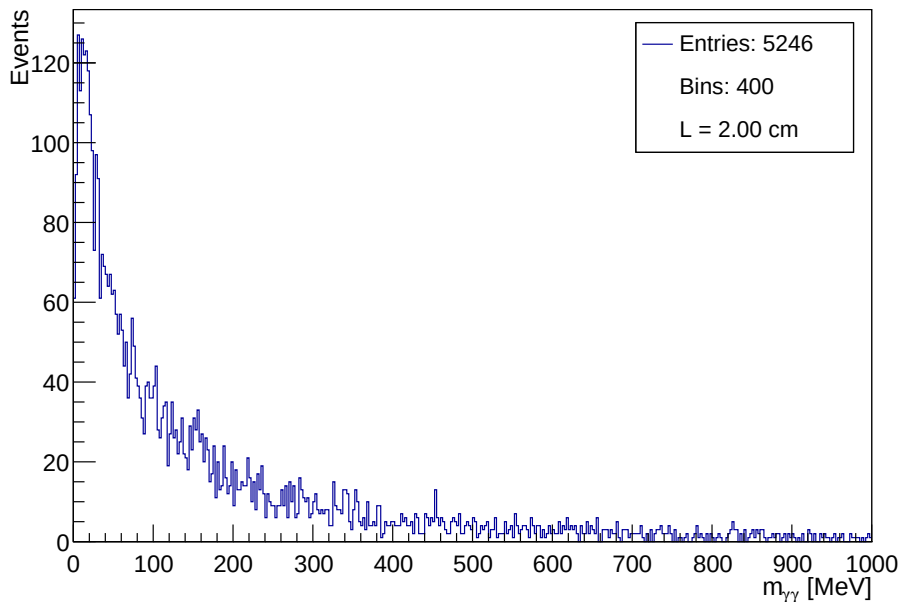
Fit π^0 peak at L=82.00 cm



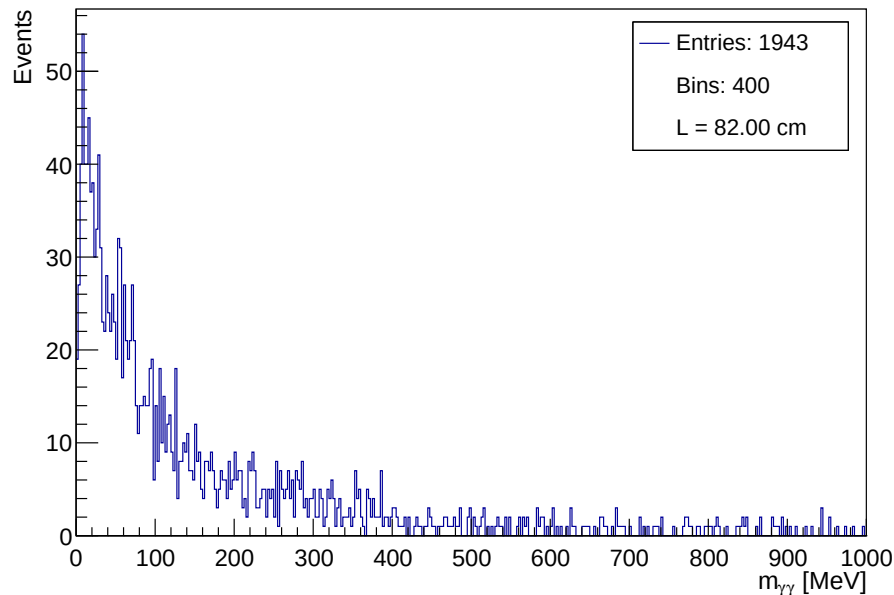
- Reconstructed **oversampled invariant-mass spectrum** for the two mesons at two representative target thicknesses.
- Total generated events: **N=10_000**
- The total number of reconstructed signal events **decreases** with increasing L
- The invariant-mass resolution **increases** with increasing L

Optimization of the target thickness

Background at $L=2.00$ cm



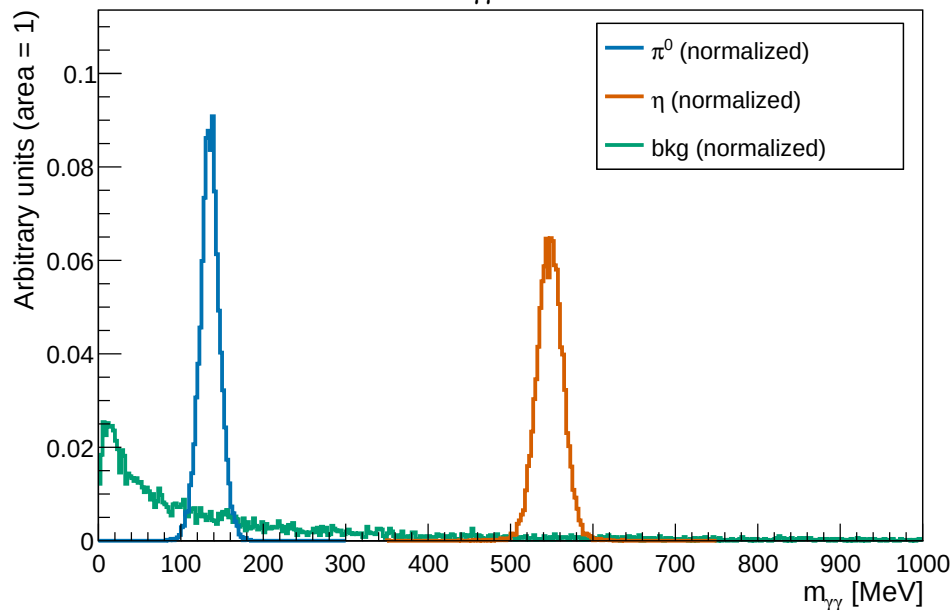
Background at $L=82.00$ cm



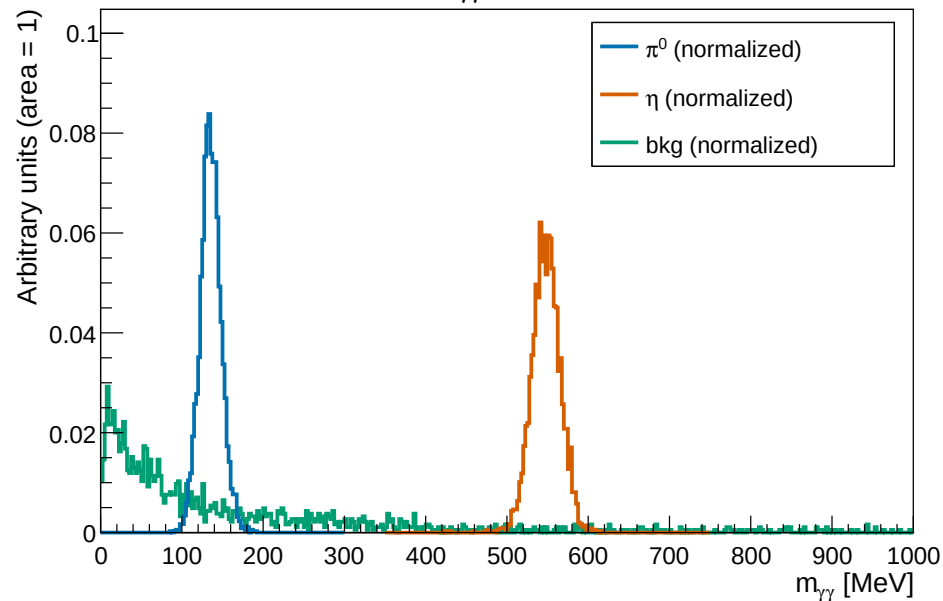
- Reconstructed **oversampled spectrum** of the **background** for two representative target thicknesses
- Total generated events: **$N=10_000$**
- **No resonant structure**
- **Soft photons** are produced with much **higher probability** than hard ones
- The total number of reconstructed signal events **decreases** with increasing L

Optimization of the target thickness

Normalized $m_{\gamma\gamma}$ at $L=2.00$ cm

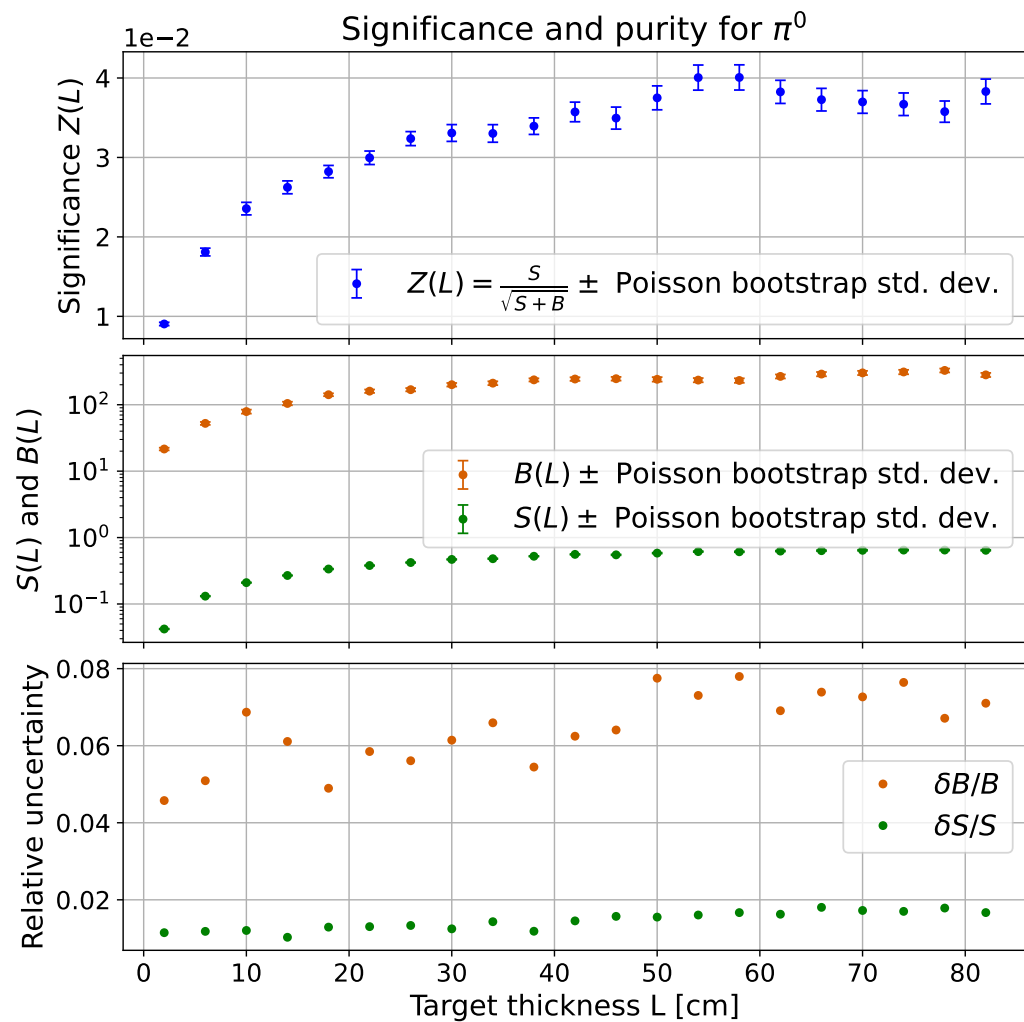
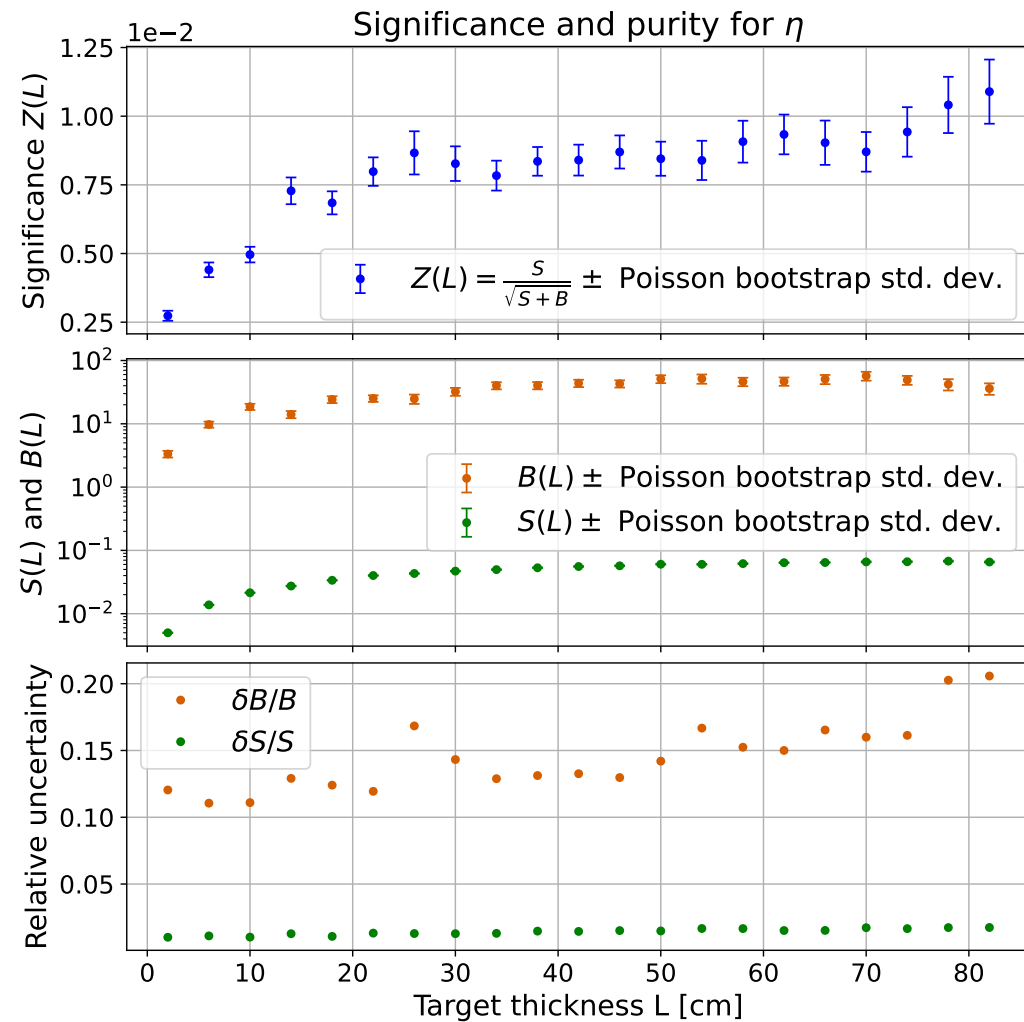


Normalized $m_{\gamma\gamma}$ at $L=82.00$ cm



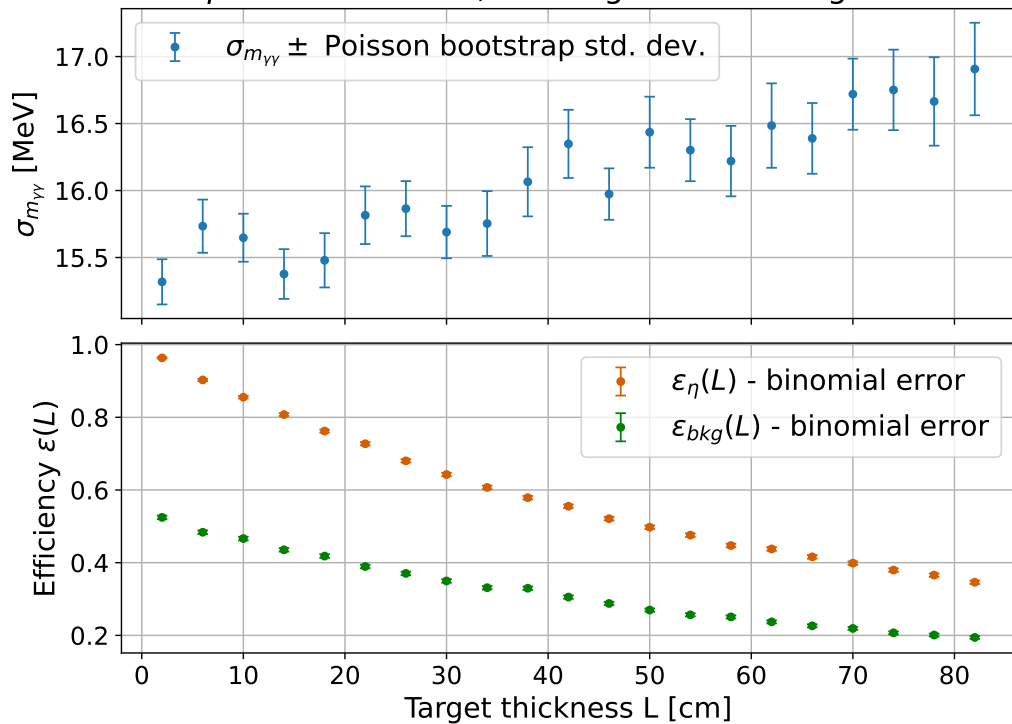
- **Normalized invariant-mass spectra** for the mesons and background
- Direct **comparison** of the **spectral shapes**
- Signals appear as **well-defined peaks** on top of a **smooth background distribution**
- **Increasing** L affects the spectra in a largely proportional way: the qualitative appearance of the normalized spectra remains similar and the difference in signal shapes **is not appreciable**

Optimization of the target thickness

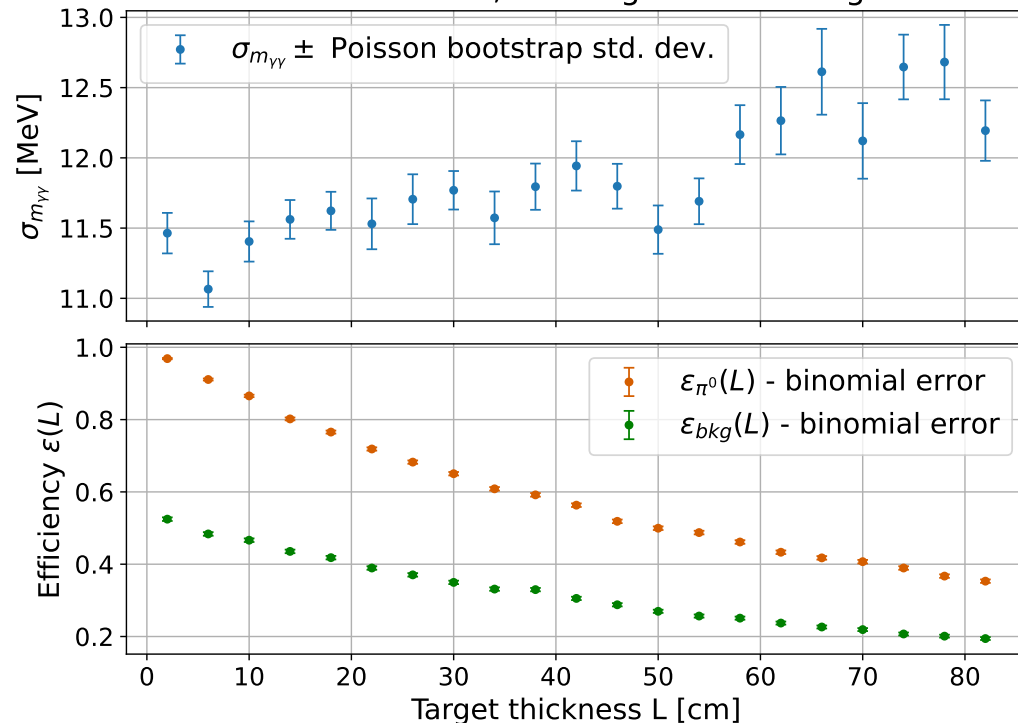


Optimization of the target thickness

η mass resolution; ϵ for signal and background



π^0 mass resolution; ϵ for signal and background



Detector efficiency:

$$\epsilon = \frac{N_{reconstructed}}{N_{generated}}$$